INSTITUTE OF TECHNOLOGY OF CAMBODIA

APPLIED MATHEMATIC AND STATISTICS

Department of Applied Mathematics and Statistics

Precision, Recall, and F1-score

Evaluating Classification Performance and Balancing Precision and Recall

Course: Advance Programming for Data Science
Group: 3

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1. Introduction

Why do we need evaluation metrics?

Classification

A model assigns an input to a category, such as positive or negative sentiment.

The evaluation problem

A model can make correct predictions and still miss important cases or produce false alarms.

Our focus

Measure the quality of positive predictions, the ability to find actual positives, and the balance between the two.

Application: classifying customer reviews as Positive or Negative.

3. Classification Evaluation

Binary sentiment classification

Input

Customer review text
Example: “The product works great.”

Model prediction

Positive or Negative
The prediction is compared with the known correct label.

Two possible classes

Positive (class 1)

The review expresses positive sentiment.

Negative (class 0)

The review expresses negative sentiment.

For this research, Positive is the designated positive class.

4. Confusion Matrix

The four outcomes used to calculate the metrics

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP) Correctly found	False Negative (FN) Missed positive
Actual Negative	False Positive (FP) False alarm	True Negative (TN) Correctly rejected

TP and TN are correct predictions. FP and FN are errors.

Precision is affected by FP. Recall is affected by FN.

5. Precision

How reliable are the positive predictions?

$$\text{Precision} = \frac{TP}{TP + FP}$$

Of all cases predicted Positive, what proportion are actually Positive?

Example

TP = 8 FP = 2
Precision = 8 / (8 + 2) = 0.80

Interpretation

80% of the model's positive predictions are correct.

High Precision means fewer false positive predictions.

6. Recall

How many actual positive cases are found?

$$\text{Recall} = \frac{TP}{TP + FN}$$

Of all cases that are actually Positive, what proportion did the model identify?

Example

TP = 8 FN = 12
Recall = 8 / (8 + 12) = 0.40

Interpretation

The model identifies 40% of all actual positive cases.

High Recall means fewer false negative predictions.

7. F1-score

A single score balancing Precision and Recall

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

F1-score is the harmonic mean of Precision and Recall.

Example

Precision = 0.80
Recall = 0.40
F1 = 2 × (0.80 × 0.40) / 1.20 = 0.533

Interpretation

F1-score = 53.3%
The low Recall pulls down the overall balance.

F1 is high only when both Precision and Recall are reasonably high.

8. Comparing the Three Metrics

Each metric answers a different question

Precision

Correctness of positive predictions

Avoid false positives (FP)

Recall

Coverage of actual positive cases

Avoid false negatives (FN)

F1-score

Balance between Precision and Recall

Useful when both matter

The best metric depends on the cost of different errors in the application.

9. Practical Applications

Two examples of why Precision and Recall matter

Example 1: Spam detection

TP = 20 FP = 5
FN = 10 TN = 65

Precision = 20 / 25 = 80%
Recall = 20 / 30 = 66.7%
F1-score = 72.7%

Interpretation: Some spam messages are missed.

Example 2: Medical screening

TP = 90 FP = 20
FN = 10 TN = 80

Precision = 90 / 110 = 81.8%
Recall = 90 / 100 = 90%
F1-score = 85.7%

Interpretation: High Recall helps reduce missed cases.

These are illustrative examples, not results from our LLM experiment.

10. Python Implementation

Using Scikit-learn to calculate the metrics

```
from sklearn.metrics import (
    precision_score, recall_score, f1_score,
    confusion_matrix
)

y_true = [1, 1, 1, 1, 0, 0, 0, 0, 0, 0]
y_pred = [1, 1, 1, 0, 1, 0, 0, 0, 0, 0]

p = precision_score(y_true, y_pred)
r = recall_score(y_true, y_pred)
f = f1_score(y_true, y_pred)

print("Precision:", p)
print("Recall:", r)
print("F1-score:", f)
print(confusion_matrix(y_true, y_pred))
```

Libraries

- Python
- Scikit-learn
- Pandas
- Matplotlib

The same functions are used to evaluate both LLMs.

Output

```
Precision: 0.75
Recall: 0.75
F1-score: 0.75
[[5 1]
 [1 3]]
```

11. Advantages and Limitations

What these metrics can and cannot tell us

Advantages

- Simple and widely used
- Precision reflects false-positive control
- Recall reflects false-negative control
- F1 combines both metrics
- Supports binary and multiclass evaluation

Limitations

- Does not measure overall language quality
- F1 does not directly include TN
- Can hide class-specific performance
- Depends on reliable ground-truth labels
- Does not account for all error costs

A high F1-score does not mean an LLM is generally intelligent or factually reliable.

12. Conclusion

Key takeaways

- Precision measures the reliability of positive predictions.
- Recall measures how many actual positives are found.
- F1-score balances Precision and Recall.
- The right metric depends on the task and the cost of errors.
- These metrics help compare LLMs on classification tasks.

Thank you